004.03 Observability Policy

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Description

This page is designed to document and explain the Observability Policy as a long-term goal and existing implementation

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What is Observability?

Observability is a measure of how well internal states of a system can be inferred from knowledge of its external outputs. Within an Observability policy, we establish a set of standards designating how we collect and manage meaningful data from our systems.

Observability vs Monitoring

Monitoring is the process of managing known systems for specific issues, such as CPU, memory, networking. Monitoring asks the same questions for known solutions and is valuable for this pattern. However, current systems , with many combinations of services, servers, containers, server-less architecture, increases the complexity of understanding present and future problems. An observability policy intends to set standards for asking different questions for discovering or detailing unknown problems or problems that monitoring simply does not look for.

How can we make a system observable?

The main component of creating observable systems is correlating the data generated within or outside of the system. Correlating a metric, like CPU usage, against a request log with the identify of a user, begins the correlation of data necessary for understanding a story that hasn't been written.

Metrics, Events, Log, Trace (MELT)

The main pillars of system observability are MELT (Metrics, Events, Log, Trace).

Alerts are considered a specific monitoring event log. Tools are required to provide a means to consolidate and use the observability data.

- Metrics: Numerical data measured over some period of time. Derived from a systems performance
- Events: Recording of an event that happened in a system. Automatically generated with a timestamp and uniquely and discretely identified
- Log: Any action on a system provided in a standardized format
- Trace: Documented record of a series of casually related events that happen on a network. List of events logs from different systems employed to full request

MELT is supported by:

- Tools: Single source for storage, execution and relation of data with real-time integration Page Status In Progress
- Alerts: A distinct event with severity or significant state change

Getting started with Logging and Monitoring

- Select useful information to share that can be correlated
- Know the metrics
  - Business Metrics - Logins, Attempts
  - Resource Metrics - CPU, Disk, Network
- Select metrics by classifying from least to most important
- Define alerts
- Classify alerts
  - Informational
  - Low
  - Medium
  - Important
  - Critical
- Control quality of log data by analyzing and adding missing information

- Contextualize event by adding
  - 1. User Identification
  - 2. Type of Event
  - 3. Date & Time
  - 4. Success/Failure
  - 5. Origination of Event
  - 6. Identify or name of Affected data, component or resource
  - 7. Unique identifier of component or resource
- Implement standardized format and content
- Centralize the log

Getting started with Observability

- How does the outside world look at my system?
- How do I ensure data from outside correlates with data on the inside?
- User requests vs CPU How do I use the data?
- Not let it be stale?
- What platforms are available for our work?